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SKU-Level Retail Demand Forecasting with Bottom-Up Aggregation Using Gradient Boosting

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Declaration of Originality

Statutory Declaration:

I hereby declare that I have developed and written the enclosed Master Thesis completely by myself and have not used sources or means without declaration in the text. I clearly marked and separately listed all the literature and all the other sources which I employed when producing this academic work, either literally or in content. I am aware that the violation of this regulation will lead to failure of the thesis.

Berlin,

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Abstract

SKU-level retail demand forecasting with bottom-up aggregation validation

Retail demand forecasting is an essential tool for inventory management and decision-making. Retail companies need accurate forecasts at different levels of aggregation, but large retail datasets often contain complex and non-linear demand patterns that are difficult to model with traditional methods.

This study investigates SKU-level retail demand forecasting with bottom-up aggregation validation using the M5 Walmart Forecasting dataset. The forecasting performance of a naïve baseline, LightGBM, and XGBoost is compared using sales, calendar, and price information together with engineered lag and rolling-average features. Forecast accuracy is commonly measured by three indicators: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Weighted Root Mean Squared Scaled Error (WRMSSE). From the output, we can see machine learning methods perform better than the naïve baseline method on almost all the metrics. Particularly, XGBoost is the method that scores the highest with MAE = 0.981, RMSE = 2.270, and WRMSSE = 0.892. Besides, by aggregating bottom-up, we verify that the total forecast is numerically unchanged at each level from Overall, through State, Store, and Category.

The findings suggest that gradient-boosting models combined with appropriate feature engineering can support practical retail demand forecasting. Future work can extend this study by adding true hierarchical reconciliation methods and probabilistic forecasting approaches.

Keywords: Retail demand Forecasting, SKU-level Forecasting, Bottom-Up Aggregation, Machine Learning, XGBoost, LightGBM, M5 Forecasting Dataset, WRMSSE

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CHAPTER 1: INTRODUCTION

1.1 Background

Retail demand forecasting might just be a tiny portion of the supply chain and inventory management puzzle; it definitely makes forecasting of retail demand a critical activity. With a good forecast, retailers are able to operate with the right stock levels reduce operational costs, limit the risk of stockouts, and actually give customers a better experience by being there for them. Still, when you have a large retail network, forecasting becomes more of a challenge because the same item sells in numerous stores, in different geographical areas, and in various product catagories. That leads to sales records that form layers of hierarchical correlations, which are hard to untangle. Until now, forecasting methods were mostly limited to techniques such as moving average, exponential smoothing, and Autoregressive Integrated Moving Average (ARIMA) for short.

We know that these methods can provide good forecasts at least for the time series that is relatively smooth but there may exist some nonlinear features and other complex demand behaviors that these methods cannot account for such as the effects of promotional campaigns, or the change of seasons. So that is the reason why more retailers will now look towards machine learning solutions for their forecasting, which might give flexibility and also a better accuracy for forecasting. Owing to the speed of machine learning's development one may already find tools that are able to catch up with such nonlinear and tangled relationships that are typical for retail sales data.

This kind of relationship modeling is more about using common sense or experience rather than being restricted to mathematical rigidness. For example, Gradient boosting methods among them Light Gradient Boosting Machine (LightGBM) and Extreme Gradient Boosting (XGBoost) have shown an excellent predictive capability on various forecasting tasks not only because of their capability to model nonlinearity but also due to their relatively high computation speed. Such models may use historical sales data in combination with calendar features, pricing data and manually created features to produce forecasts which are more accurate. A retail company needs to produce the forecast not only at the level of each SKU but also at the departments, product categories, store, state, and company levels when it wants to understand the overall performance and potential of the company, especially for the large ones.

The problem is that if forecasts are made on different levels without taking the whole hierarchy into account, they can contradict each other as a result, lower-level results don't match the forecast at higher levels. That discrepancy can interfere with the production planning, inventory distribution, and other business decisions.

This is solved using hierarchical forecasting which links forecast data on one level with another so the numbers are consistent throughout the hierarchy. Hierarchical forecasting considers the interdependence of various parts and does

not treat individual time series as independent entities. Decision-makers are provided more cohesive sets of forecast data which are helpful for operational planning as well as for strategic planning.

The M5 dataset made public by the US retail giant Walmart and available at Kaggle is a popular forecasting challenge dataset, being used frequently for the purpose of retail demand forecasting research. Sales data of different products across a variety of stores and states from Walmart is available from the dataset over several years on the daily basis.

In addition to historical sales numbers, the dataset comprises information of the calendar, price, and special event indicators and therefore, it is quite suited for modern forecasting approaches testing.

This paper aims at looking into the issue of demand for products sold in a retail environment at different levels that are structured hierarchically and using these demand forecasts to carry out operations that include inventory management through comparison of the traditional approach of forecasting against two machine learning based forecasting ones LightGBM and XGBoost based on the M5 retail sales dataset. The forecast results for each forecast method are measured in terms of Mean Absolute Error (MAE), Root Mean Squared error (RMSE) and Weighted root Mean Scaled Squared Error (WRMSSE), This metric is also the official evaluation metric in the M5 competition. Besides, the validation is conducted through verifying the consistency at all levels of the retail hierarchy (hierarchical

aggregation).

1.2 Problem Statement

Besides the shifting nature of consumer buying habits seasonality, promotional campaigns, holidays, price variations and lots of other external factors make a major contribution to demand variability. These factors create highly diverse demand patterns which are difficult to grasp if traditional statistical forecasting is used.

Another significant challenge is ensuring coherence of forecasts at different forecasting levels in hierarchies, which kind of is like an alignment problem even if the data is in flux. Retailers rely on multi-level forecasts which, for example, could be overall company level sales forecasts down to very granular level forecasts for individual SKUs. If forecasting is carried out separately for each level inconsistency is likely to happen because total sales based on lower level forecasts may not be in line with those at the higher levels. Deviations of this nature make it tough to control stock and plan operations.

Although transformer-based deep learning models have recently demonstrated promising performance in time-series forecasting, they typically require long historical sequences for each individual time series. In the M5 dataset, many SKU-store combinations contain relatively short or sparse sales histories, making

transformer-based models unsuitable for reliable training at the lowest hierarchy level. Therefore, there is a need to investigate practical machine learning approaches capable of producing accurate retail demand forecasts while maintaining consistency across hierarchical aggregation levels. This research addresses this need by comparing LightGBM and XGBoost with a naive forecasting baseline using the M5 retail dataset.

1.3 Research Aim

Through this research, we hope to develop and assess a sku-based retail demand forecasting method which will make use of machine learning techniques and validation of bottom-up aggregation on the M5 Forecasting dataset

1.4 Research Objectives

The objectives of this study are:

1) Preprocess and integrate the M5 retail sales sales, calendar, and pricing data to prepare SKU-level forecast demand dataset.
2) Generate forecasting features that capture historical pattern of demand including lag, and rolling variables in order to enhance prediction accuracy.
3) Use LightGBM and XGBoost models for retail demand forecasting at SKU level and evaluate their performance along with a naive forecasting baseline comparison.
4) Assess forecasting accuracy through MAE, RMSE, and WRMSSE-based error metrics.
5) Make sure that the numerical consistency of the SKU-level forecasts is not violated after bottom-up aggregation among the different levels of retail hierarchy.
6) Analyze and contrast the forecasting results of the tested machine learning models, and reveal their real-world implications for demand forecasting in retail.

1.5 Research Questions

This study attempts to provide an answer to a set of research questions, a task relatively easy to imagine but a matter of great complexity when it comes to details.

RQ1: Do machine learning models really help improve retail demand forecasting

accuracy, if we compare them to a naive baseline of forecasting which does not include any advanced algorithms, in a direct head-to-head manner?

RQ2: Among machine learning models used for forecasting on the M5 retail dataset, which performs best overall?

RQ3: Does bottom-up aggregation preserve numerical consistency across hierarchy levels?

1.6 Scope of the Study

This research presents a SKU-level retail demand forecast based on the Walmart dataset for M5 Forecasting. The study combines historical sales, calendar, and price data from Walmart with some engineered features like lag variables and rolling statistics to build the forecast models. Three forecasting techniques that are being studied are a naive forecasting baseline, LightGBM, and XGBoost.

The quality of forecasting is measured by Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Weighted Root Mean Squared Scaled Error (WRMSSE)based evaluation. Apart from model assessment, the research also uses a bottom-up aggregation method to test the numerical consistency between SKU-level forecasts and aggregated forecasts at different retail hierarchy levels such as Overall, State, Store, and Category.

This article is mainly concerned with the point forecasts for the SKU level and the checking the numerical consistency of forecasting after bottom-up aggregation. No implementation of the Minimum Trace (MinT) method is mentioned for reconciling forecasts at different levels of the hierarchy through changes and modifications etc. They are viewed as research options for the future.

The Temporal Fusion Transformer (TFT) method was identified as a potential forecasting method due to the ability of the model to capture and represent complex relationships in data across time. Nevertheless, the TFT forecasting method was not included into the final comparison as the research focus primarily targeted machine learning tree-based models that can be used efficiently at scale for the purpose of the large-scale retail demand forecasting. For that matter, the main experiment was about comparing the performance of the naive forecasting baseline, LightGBM, and XGBoost.

1.7 Significance of the Study

Using the M5 Forecasting dataset, this paper assesses, from an actual point of view, different machine learning models for predicting the demand of individual product items. The research compares LightGBM and XGBoost with a naïve forecasting baseline in the light of the features crafted from the past sales data, calendar events, and pricing data. The study shows the merits and the efficiency of feature-engineering techniques coupled with gradient boosting in order to forecast the demand with large-scale retail datasets.

The research also reveals an innovative method of bottom-up aggregation validation that the paper uses in order to determine the numerical coherence of the forecasts of each product in a retailer hierarchy. It helps understand the ways forecasts made for items can be summed up to provide analysis at different organisational levels. Unfortunately, hierarchical reconciliation techniques are

not discussed at all which opens up areas where future studies can make a significant contribution.

Practically, the research may assist manufacturers and retailers at a store level in better demand planning, maintaining an optimal inventory, making better supply decisions etc. The research also demonstrates how scalable tree-based machine learning models can be used for retail datasets that show complex and probably sparse demand patterns. Apart from that, the research touches base with the challenges of implementing more computing-consuming deep learning methods, Temporal Fusion Transformer included, on retail time series with limited historical observations.

1.8 Chapter Summary

This first chapter conveyed the background of the study being making a simple background information about what this study was all about, its aims, research questions, deliminations and significance of this study. It set the stage to motivate machine learning techniques applied to hierarchical retail forecasting and it indicated the broad plan for this research. The next chapter will review literature related to areas that are pertinent to this research: the field of retail demand forecasting, hierarchical forecasting methods, machine learning techniques and evaluation metrics.

CHAPTER 2: Literature Review

2.1 Introduction to Retail Demand Forecasting

Demand forecasting forms a critical component of a supply chain and inventory management, providing retailers with a means to make projections on customers' future demand. The role of demand forecasting extends beyond operational areas such as inventory planning, procurement, workforce scheduling and transportation management to include enabling retailers to optimise their warehouse operations too. With precise demand forecasting, companies can be assured of stock availability, limit the cost of goods in stock while also keeping the stock short risk minimal. In view of this, retail forecasting is widely regarded as one of the crucial functions in modern retail business.

Demand forecasting in the retail industry may be a particularly difficult job as purchasing behaviors of consumers are determined by a host of internal and external variables. The factors that drive demand can be, for instance, the changing of seasons, the running of promotions, being on holiday, the way products are priced, the conditions of the economy, and the changes in trends of consumers. The combination of different influences is generally so perplexing that forecasting demand for goods becomes quite a daunting task. Forecast mistake could mean the retailers carrying excessive goods, which in turn would require more space for storage, product discounts, and finally customer dissatisfaction from stockouts would be the cause of losing sales.

Today's retail firms have different operational layers starting from the organisation level, regional, local, product category, department, down to the individual Stock Keeping Unit (SKU) level. The forecasts vary significantly with the management level, the degree to which data is summarised. In this case, store managers can only rely on SKU-level forecasting forecasts for restocking decision making; on the contrary, higher-level forecasters can make use of aggregated forecasts as the basis for the budgeting activities, allocating resources, and strategic planning. So, forecasting for the store should involve both detailed product-level demand and summarised business-level data for effective decision making.

Historically, forecasting retail demand was conducted through statistical, time series based methods such as the Moving Average, Exponential Smoothing, and Autoregressive Integrated Moving Average (ARIMA). These are great when demand patterns don't change a lot, and historical data is enough to predict for some time. Nevertheless, these retail demand patterns, with many changes such as promotions, pricing changes, holiday related events etc., tend to violate the assumptions of these traditional statistical models. As the size of data increases, the accuracy of such traditional methods is likely to suffer as well.

Retain data growth has pushed machine learning to be used in demand forecasting. Machine learning algorithms differ from traditional statistical methods as they extract complex and potentially non-linear relationships from historical

data without any strong assumption about the data distribution. They can easily incorporate many variables at once, such as old-time sales, price detail, calendar features, promotional activities and even new variables constructed from previous demand. This versatility makes machine learning a good choice for big retail demand predictions tasks. Out of various machine learning algorithms, Gradient Boosting has been found to have superior forecasting ability for retail structured data. LightGBM and XGBoost are able to process big data and uncover complicated correlations of explanatory variables with product demand very well. Besides these tree-based machine learning models, Deep Learning methods including RNN, LSTM, and TFT have also had their share in retail forecasting. Although results have been good with these models, they mostly call for a lot of historical data besides heavy resources, which may hinder the deployment of deep learning models to the demand prediction of sparse SKU level series.

Another important development in retail forecasting is hierarchical forecasting, which considers demand across multiple aggregation levels within an organisation. Instead of analysing a single time series in isolation, hierarchical forecasting recognises the relationships between detailed and aggregated demand series. This perspective is particularly valuable in retail environments where operational decisions are made at both product and organisational levels.

Building upon these developments, this research investigates SKU-level retail demand forecasting using the M5 Forecasting dataset. The study evaluates the forecasting performance of LightGBM and XGBoost against a Naïve Forecast baseline using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Weighted Root Mean Squared Scaled Error (WRMSSE). Furthermore, SKU-level forecasts are aggregated using a bottom-up aggregation approach across selected organisational levels to validate the numerical consistency between detailed and aggregated forecasts. The study therefore focuses on evaluating the effectiveness of machine learning models for SKU-level retail demand forecasting while examining the consistency of aggregated forecasts within the retail hierarchy.

2.2 Hierarchical Forecasting

Hierarchical forecasting is an important forecasting approach for datasets organised into multiple levels of aggregation. It is widely applied in sectors such as retail, manufacturing, healthcare, and energy management, where forecasts are required at different organisational levels rather than for a single time series. In retail, for example, organisations analyse demand at the company, state, store, product category, department, and Stock Keeping Unit (SKU) levels. These levels form a hierarchy in which lower-level observations are aggregated to produce higher-level totals. Consequently, forecasts generated across different hierarchical levels should remain numerically consistent to support effective decision-making throughout the organisation.

Within a retail hierarchy, total company sales represent the highest aggregation level, followed by state, store, category, department, and individual SKU sales. Since higher-level values are obtained by summing lower-level observations, independently generated forecasts may become inconsistent. For instance, the total forecast at the store level may not necessarily be the same as that of the state level forecast. Such inconsistencies have the potential to affect negatively inventory management, budgeting, logistics, and strategic planning. Consistency of forecasts at all the aggregation levels forms an integral part of hierarchical forecasting. However, forecasting errors at the SKU level may accumulate during aggregation.

Hierarchical forecasting seeks to achieve logical consistency of forecasts at all levels. Coherent forecasts ensure that lower-level predictions aggregate correctly to higher organisational levels. This is particularly important because different business functions rely on forecasts at different levels. Operational teams need precise SKU forecasts for replenishing stocks and managing inventories, while the management needs aggregate forecasts for financial budgeting and resource planning. Employing incompatible forecasts within different departments can result in ineffective decision-making and operations. Although MinT has demonstrated strong performance in previous research, its implementation was beyond the scope of the present study.

Several reconciliation approaches have been proposed to maintain forecast consistency. The most widely used methods include Bottom-Up, Top-Down, Middle-Out, and Minimum Trace (MinT) reconciliation.

Bottom-Up Approach: This forecasting technique makes forecasts from the bottom of the hierarchy, which usually refers to the SKU level, and thereafter aggregates the forecasts to create forecasts at department, category, store, regional, and organizational levels. This technique ensures that details of product information are not lost, and coherent forecasts are always achieved. However, its accuracy may decrease when individual products exhibit highly intermittent demand or large random fluctuations, as forecasting errors at the lowest level accumulate during aggregation.

The Top-Down technique employs the reverse approach whereby forecasting starts at the higher level, for instance, organisational total sales, and then distributing the forecasted values at the lower level based on ratios. Forecasting demand at the higher level is usually less volatile than forecasting demand at the product level. However, the drawback of this technique is that it fails to account for local and unique behaviour of products within the hierarchical structure.

The middle-out method is a blend between Bottom-Up and Top-Down forecasting where the forecasts are made at the middle level, e.g., product categories or stores, before the forecasting results are summed and disseminated. The underlying idea is to strike a balance between very detailed forecasting on one hand and, on the other hand, the stability of the aggregated demand. Still, the choice of the right intermediate level is a dataset dependent thing and poses a challenge

in its implementation.

Lately, statistical methods of reconciling hierarchical data sets have been attracting the limelight. Among them, Minimum Trace (MinT) is considered a very prominent one. MinT operates differently; it forecasts separately at each level of the hierarchy and then, through reconciliation, minimizes an overall forecast error covariance. It keeps the forecast coherent as all the reconciliation constraints are included in the minimisation problem. MinT takes more into the consideration of various levels of aggregation simultaneously so it is not rare for MinT to deliver more precise coherent forecasts over those of conventional Bottom-Up or Top-Down methods. Still, the estimation of the covariance matrices of very sizeable dimensions leads to a rise in computational cost and time, especially, with retail datasets comprising thousands of products.

More and more recent studies have combined hierarchical forecasting and machine learning techniques like LightGBM or XGBoost. They have the effect of giving more accurate predictions and, at the same time, maintaining consistency by using hierarchical aggregation across different levels within a company. This synergy leads to a robust and scalable framework, particularly useful for forecasting retail demand on a very large basis.

The M5 Forecasting Competition made great strides in hierarchical forecasting research by introducing a large-scale retail dataset which the participants arranged across several aggregation levels. This not only brought the attention of the forecasting community to work on hierarchical consistency along with producing high-accuracy predictions but also resulted in adopting the Weighted Root Mean Squared Scaled Error (WRMSSE) as the sole metric for evaluating forecasting performance. This metric, by being the only one, is a very effective tool, as forecasting evaluation can be performed not just at some aggregation level of the hierarchy but at the whole hierarchy level. WRMSSE has become the standard evaluation measure in hierarchically retail forecasting studies.

Our research work is centered around the use of hierarchical forecasts as one of the building blocks of the proposed forecasting method. Firstly, products' demand is forecasted using machine learning models and then these forecasts are further grouped at the hierarchy level to see if the aggregated data match the hierarchy. It was confirmed that, through different hierarchy levels, aggregated sales remained exactly the same. This indicated that the aggregation process was performed accurately without any loss or duplication of data. Literature review was undertaken where advanced reconciliation methods like MinT that were considered but not used since the study was primarily aimed at comparing models and, therefore, hierarchical consistency verification is of much importance to this research. The study of reconciliation advanced methods can serve as a new research avenue.

The application of hierarchical demand forecasting is a great decision making tool at different organizational levels of the retail industry. The main advantage of the approach is that it creates a consistent link between forecasts at the finest level

and those aggregated forecasts. Given that retails data are getting more extensive and diverse, a combination of ml methods and hierarchical demand forecasting concepts shall remain the major direction of academic and industry-level works for quite some time.

2.3 Machine Learning for Retail Demand Forecasting

New technologies in data gathering have altered the way of forecast demand in retail. Since statistical methods are relatively simple and explainable they have been the dominant technique, for many years. However, as the retail environment has become more complex and unpredictable and the available data has become more voluminous, machine learning methods have been gradually increasing in popularity. They are able to uncover the sales hidden patterns. Compared with traditional methods based on fixed assumptions, machine learning algorithms learn to understand the relationship among the data. Consequently, they are able to capture the interactions among factors.

Accurate forecasting is not an easy task. Demand depends on a variety of different variables at any given moment. One variable may unexpectedly change which could have a significant impact in a direction while another does exactly the opposite. For example, in retail business customer purchasing behavior is affected by various elements such as seasonal patterns, promotions or holidays, weather, product price, etc. Additionally, some of the major market events are also a factor. These aspects, however, have complex interactions which are very challenging to model with conventional time series forecasting techniques. Also, the major characteristic of retail transactions to consider is the possibility of having a whole bunch of demand records simultaneously. Some of the records could be missing while different product lines and stores can vary in a big way on the magnitude. For such problems, machine learning techniques, on the other hand, have a much better fit. They can train on large scale data and discover multiple-layered patterns and complex interactions.

Traditional models of predictive systems before machine learning were based only on a narrow group of very simple methods: Exponential smoothing, Moving averages, Autocorrelative model. In fact, in a case of pure demand unit, the effectiveness of such models largely is determined by the availability of certain conditions and a favorable environment. They should, if anything, have a good foundation to operate on fully satisfactory. One condition is whether there's any change to the forecasted demand based upon previous periods; if the average demand has not changed for at least three consecutive months, then the sales projections will likely remain the same as long as each of the important underlying assumptions associated with the forecasting method remain the same (linearity and time stationarity). As demand becomes increasingly driven by factors that are outside of a retailer's control (e.g., promotions, holidays, stock availability, shifts in consumer trends), it is likely that many of conventional forecasting approaches will cease being of value for producing accurate predictions. Under conditions traditional statistical methods may struggle to adapt quickly. This

results in reduced forecasting accuracy.

Machine learning techniques are able to address many of these limitations by being able to learn from past observations instead of using pre-defined mathematical assumptions. When building the model, algorithms will examine past sales data, and any other variables that may provide an insight into future sales, to find hidden relationships that impact the demand for products in the future. Historical sales lags, moving averages, past and current product prices, day-of-the-week effects, seasons, and promotional campaigns are among the variables which could be used. Through simultaneous use of several explanatory variables, it is possible for machine learning techniques to pick up from the data the interplay effects which cannot be easily identified by traditional methods.

Decision Trees were one of the earliest used machine learning techniques for solving the problem of forecasting. A Decision Tree segments the dataset (or population) on the basis of one of the features' values so that it finds the greatest information gain or least predicted error by doing this recursively until there are no further opportunities for segmentation in the dataset. Although Decision Trees are relatively easy to understand and do not require a lot of data preparation, a major drawback of using a Decision Tree for forecasting is that they often result in an overfitted model due to excessive depth of the tree. Consequently, individual Decision Trees generally provide limited forecasting accuracy for complex retail datasets.

Although Decision Trees are simple to understand and easy to interpret, they often suffer from overfitting when applied to large and complex retail datasets. Small changes in the training data can produce very different tree structures, resulting in unstable forecasts. Consequently, researchers introduced ensemble learning techniques to improve forecasting accuracy by combining the predictions of multiple decision trees. Ensemble methods generally provide better generalisation and produce more reliable forecasts than a single Decision Tree, particularly for large-scale retail demand forecasting.

One of the earliest and most widely adopted ensemble learning techniques is the Random Forest algorithm. Random Forest constructs a large number of independent Decision Trees using randomly selected subsets of observations and predictor variables. The final prediction is obtained by averaging the outputs of all individual trees. This approach reduces the risk of overfitting and improves prediction stability compared with a single Decision Tree. Random Forest has been successfully applied in various forecasting applications because of its robustness and ability to model nonlinear relationships. However, since each tree is trained independently, Random Forest cannot sequentially correct forecasting errors during training. This limitation motivated the development of boosting-based algorithms.

Gradient Boosting brought about a fundamentally different idea of how the learning process is carried out with the help of Decision Trees. Instead of building them independently, Gradient Boosting sequentially creates Decision Trees. In

each round, a new tree has to be grown such that it will be good at fixing the errors left by the previous trees. Therefore, Gradient Boosting is able to achieve excellent performance and uncover complicated non-linear patterns hidden in the data gradually through many rounds of error-correction. While a standard implementation of Gradient Boosting is able to yield pretty good prediction results, a drawback is that it will require a lot of computing power if large retail data sets with thousands of products and millions of sales transactions are going to be used. Computational limitations have prompted the development of several improved variants of Gradient Boosting. LightGBM and XGBoost are the two most popular algorithms for predicting retail demands these days. On the original Gradient Boosting setup, both methods add value while keeping high forecasting accuracy relatively. LightGBM mainly concentrates on speed via histogram based learning and leaf-wise tree growth. This feature allows faster model development using data sets of very large size.

Conversely, XGBoost relies on thorough optimisation through regularisation and the use of second-order gradients, thus making the model more stable and reducing the overfitting danger. Most published studies have pointed out the superiority of the two algorithms over the other traditional statistical forecasting methods in the case of structured retail data. The latest developments in deep learning, have also presented a whole array of forecasting models which are able to learn very complicated temporal relationships directly from sequential data. Neural Network (ANN) architectures, Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRUs), as well as Transformer-based models, have all displayed excellent forecasting skills and have been employed in many different sectors. Tree-based machine learning algorithms are generally different in a way they work: unlike them deep learning systems automatically create feature representations directly from the historical record without requiring extensive manual feature engineering. Still, as a downside, they usually call for a large quantity of high-quality training data and a lot of computing power.

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The latest developments in deep learning, have also presented a whole array of forecasting models which are able to learn very complicated temporal relationships directly from sequential data. Neural Network (ANN) architectures, Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRUs), as well as Transformer-based models, have all displayed excellent forecasting skills and have been employed in many different sectors. Tree-based machine learning algorithms are generally different in a way they work: unlike them deep learning systems automatically create feature representations directly from the historical record without requiring extensive manual feature engineering. Still, as a downside, they usually call for a large quantity of high-quality training data and a lot of computing power.

Among recent deep learning models, the Temporal Fusion Transformer (TFT) has received considerable attention because of its ability to combine recurrent neural networks with attention mechanisms while simultaneously handling multiple input variables. TFT has demonstrated excellent forecasting performance in several multivariate time-series applications and provides improved interpretability through variable selection and attention layers. Nevertheless, the model performs best when long historical sequences are available for each individual time series. Many SKU-level series within the M5 dataset contain sparse historical observations, making reliable TFT training difficult. For this reason, TFT was reviewed as part of the literature but was not included in the final experimental comparison conducted in this study.

Feature engineering remains one of the most important components of machine learning-based retail forecasting. Instead of relying only on raw historical sales values, researchers generate additional explanatory variables such as lag features, rolling averages, moving statistics, calendar variables, selling prices, promotional indicators, and holiday information. These engineered features provide valuable information about temporal demand behaviour and enable machine learning algorithms to identify complex nonlinear relationships more effectively, leading to improved forecasting accuracy.

The importance of combining feature engineering with advanced machine learning methods was clearly demonstrated during the M5 Forecasting Competition. Many of the highest-ranked forecasting solutions relied primarily on gradient boosting algorithms, particularly LightGBM and XGBoost, together with extensive feature engineering techniques. The competition also introduced the Weighted Root Mean Squared Scaled Error (WRMSSE) as the official evaluation metric, highlighting the importance of assessing forecasting performance

across multiple hierarchical aggregation levels rather than using only traditional accuracy measures.

Based on the literature reviewed, machine learning methods provide substantial advantages over traditional statistical forecasting approaches for large-scale retail demand forecasting. Among the available algorithms, LightGBM and XGBoost offer an effective balance between forecasting accuracy, computational efficiency, scalability, and practical implementation. Consequently, these two algorithms were selected as the primary forecasting models evaluated in this research, while the Naïve Forecast was used as a baseline for performance comparison.

2.4 Light Gradient Boosting Machine (LightGBM)

Light Gradient Boosting Machine (LightGBM) is a gradient boosting framework developed by Microsoft that is designed for efficient and scalable machine learning on structured datasets. Like other gradient boosting algorithms, LightGBM builds an ensemble of decision trees sequentially, where each new tree attempts to reduce the prediction errors made by the previous trees. This recursive learning method helps the model to discover complex links between the predictor and the target variables and, at the same time, keep the model prediction accuracy at a high level (Ke et al., 2017).

A distinctive feature of LightGBM compared to classical gradient boosting approaches is leaf-wise tree growth. Not all nodes are being expanded to a certain depth (level-wise growth) at once. In this way, the leaf with the biggest prediction error reduction will get the next split. In general, the algorithm ends up with more complex trees, and, with fewer splits, the same high level of performance may be maintained. However, one has to fine tune the parameters carefully least one run into overfitting. Moreover, a histogram learning mechanism is applied to the algorithm which transforms continuous feature values into discrete categories before model training. As a result, both memory and processing resources are saved without compromising the predictive performance significantly.

LightGBM also makes use of a technique called Gradient-based One-Side Sampling (GOSS), and an Exclusive Feature Bundling (EFB) method for computational speed gains. GOSS retains observations with large prediction errors while randomly sampling observations with smaller gradients, allowing the algorithm to focus on the most informative training examples. EFB combines mutually exclusive sparse features into a smaller number of variables, reducing dataset dimensionality without substantial information loss. These optimisation techniques make LightGBM particularly suitable for large-scale retail datasets containing numerous products, stores, calendar variables and pricing information.

LightGBM has been widely applied in retail demand forecasting because it can effectively model interactions among historical sales, calendar effects, prices, promotional events and other explanatory variables. Previous studies have shown that the algorithm performs well on large, structured datasets while requiring relatively short training times compared with many alternative machine learning

methods. However, because LightGBM is an ensemble of many decision trees, its predictions are generally less interpretable than those produced by traditional statistical forecasting models, and careful hyperparameter tuning is required to minimise overfitting (Ke et al., 2017).

In this research, LightGBM was selected because it provides a suitable balance between forecasting accuracy and computational efficiency for large-scale retail demand forecasting. The model was trained using engineered features derived from historical sales, including lag variables, rolling statistics, calendar information and selling prices. Its forecasting performance was evaluated against the Naïve Forecast baseline and the XGBoost model using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Weighted Root Mean Squared Scaled Error (WRMSSE).

2.5 Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting or XGBoost is a very efficient implementation of gradient boosting which is now commonly adopted for different data problems including structured data for regression, classification and forecasting (Chen & Guestrin, 2016). Basically the same as other gradient boosting methods, XGBoost builds an ensemble of decision trees in a sequential manner, and each tree is made to find out what the previous tree could not so that the prediction errors get reduced.

Among several characteristics that differentiate XGBoost from a typical gradient boosting algorithm, one is second-order gradient optimisation and other ones are using L1 and L2 regularisations to reduce overfitting, supporting parallel processing, and the ability to automatically manage missing data during the construction of the tree. In this way, both the model robustness and the speed of training are enhanced, making XGBoost a good choice for large retail forecasting datasets that contain hundreds of explanatory variables.

Previous studies have successfully applied XGBoost to retail demand forecasting because it can model complex non-linear relationships between historical sales, prices, promotional activities, calendar effects, and seasonal patterns. The algorithm has consistently demonstrated competitive forecasting performance in structured tabular datasets and has therefore become a common benchmark in retail forecasting research (Chen Guestrin, 2016).

In this study, XGBoost was selected because it is well suited to the engineered features developed from the M5 retail dataset, including lag variables, rolling statistics, calendar features and selling prices. Its forecasting performance was evaluated alongside the Naïve Forecast baseline and LightGBM using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Weighted Root Mean Squared Scaled Error (WRMSSE). Rather than assuming superiority, the comparative evaluation presented in Chapter 4 determines the relative forecasting performance of the three models.

2.6 Temporal Fusion Transformer (TFT)

TFT (Temporally Fusion Transformer Lim et al. (2021)) is a complex deep learning solution designed to multivariate time-Series forecasting. It combines recurrent neural networks, attention mechanisms, and variable selection networks to identify complex time relationships while offering interpretability level.

TFT was found to provide good forecasting power in a variety of fields like energy demand forecasting, traffic prediction, financial forecasting, and retail demand forecasting, according to some papers. Unfortunately, the model has a huge amount of parameters and generally needs powerful computing equipment along with a well-prepared sequential training dataset. Thus, the realization of a project relies not only on the amount of historical data at hand but also on the available computational power."

Even though the TFT model was considered as one of the candidates at the research planning stage, it was abandoned during the final experiment stage. This work intended primarily to assess learning techniques compatible with computing resources and development time limits. Hence, the experimental analysis centered mainly on Naive Forecast baseline, LightGBM, and XGBoost, which have demonstrated promising results in retail forecasting with structured data.

However, the TFT approach is going to be very useful to study and understand forecasting models through deep learning techniques further. In an upcoming research work if there will be a high amount of computational resources and a study design tailored for deep learning models, it may be possible to test TFT alongside tree-based machine learning models and evaluate if deep machine learning and hierarchical forecast reconciliation together result in more accurate forecasts across the forecasted aggregation levels."

2.7 Research Gap and Chapter Summary

This chapter discussed the existing body of work related to retail demand forecasting and the machine learning techniques used in this research. It commenced by tracing the change in demandforecasting methods from conventional statistical methods, such as Moving Averages, Exponential Smoothing, and ARIMA, to newer machine learning approaches that can effectively capture complex nonlinear relationships in structured retail data. The chapter highlighted a category of machine learning models, especially LightGBM and XGBoost decision tree models, which have demonstrated superior predictive performance as well as computational efficiency, and thus have become the preferred choice of methods in retail forecasting.

Furthermore. This part dealt with recent deep learning developments, with focus on the Temporal Fusion Transformer (TFT) model. Although several applications have shown that TFT is quite accurate for multivariate forecasting, this research paper did not implement TFT practically as one of the limitations

of this study, it focused mainly on tree-based machine learning algorithms that fit well with the available computational power and project size.

The paper also acknowledged the significance of hierarchical forecasting in retail supply chains, the scenario where forecasts must be done at various levels of detail. The review of existing literature showed that consistency across organizational levels is not only desirable but also critical for effective inventory planning and decision-making. In addition, several forecasting evaluation techniques were covered such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the WRMSSE, which is the commonly used metric for forecasting performance in the M5 Retail dataset.

A number of research directions were identified during the literature review that serve as motivations for the current study. While machine learning techniques were widely exploited for prediction in retail markets, much fewer studies focused on retail hierarchy performance evaluation using a combination of feature engineering and various forecast error evaluation metrics at the same time. In the present work, the comparative forecasting performance of Naïve Forecast baseline, LightGBM and XGBoost is explored through the use of manually crafted retail features that are based on sales history, calendar data and selling prices. The detailed procedure explaining how different forecasting models were developed, trained and evaluated is the content of the next chapter.

Chapter 3 – Research Methodology

3.1 Research Design

The research team chose quantitative research as the dominant method here. Since the study aim was all about numbers i.e., numerical analysis of the retail sales data, forecasting model development, and accuracy comparison through standard performance criteria quantitative approach suited it very well. The experimental study method was implemented, so that all different forecasting models could be trained on the same data, and tested under same conditions, resulting in comparison that makes fair sense.

Walmart stores from different parts of the U.S. supplied historical daily sales information to create the M5 Forecasting Accuracy dataset which the research relied on. The M5 dataset provides sales data at Stock Keeping Unit (SKU) level, calendar data, and weekly product prices. A complete forecasting dataset appropriate for running machine learning models in a supervised way was compiled by integrating these separate data sources. This dataset is realistic at representing the retail environment and it is probably the largest demand forecasting benchmarking database in research nowadays.

The entire research work was made up of a number of different steps carried out one after the other. First of all, the three original M5 datasets were connected by using common identifiers, and temporal attributes, resulting in a single analytical dataset. Then data purification steps were undertaken to eliminate inconsistencies, fill missing values, and get the data into the right shape for machine learning algorithms. After this step, additional explanatory variables were created by employing feature engineering strategies such as lag features, rolling averages, and calendar-based variables meant to represent the temporal purchasing patterns.

Once the data had been prepared, it was split into a training and a test set. Historical observations were used for training the model, and the prediction of the last twenty-eight days of sales was done out-of-sample, and so, was the performance evaluation. This sort of time series split is quite typical and reflects real-life situations where one has to predict demand based only on past data.

Three forecasting methods were developed and tested in this project. The naive forecast was used as a benchmark, a simple assumption of the sales being identical to the latest few days' sales. Then, two machine learning techniques, namely, the Light Gradient Boosting Machine (LightGBM) and the eXtreme Gradient Boosting (XGBoost), were used with the features created for the prediction. The choice of these ML models was based on the fact that many studies have reported the efficiency of these algorithms for structured retail forecasting, and they also keep up with the computational efficiency requirements.

Also, we tried Temporal Fusion Transformer (TFT) on a small basis during our work. Unfortunately, many of the SKU-store combinations did not have sufficient

historical data to do training by means of deep learning with reasonable stability. Because of this exclusivity criterion, we do not take the transformer model results into account as we believe they would only be too complex for the data features available. Choosing only the forecasting models that could realistically be expected to perform well, instead of ones which are mainly theoretical, would lead to more reliable modeling results given the amount and kind of the data available for analysis.

After training the models, the accuracy of forecasting was measured using three complementary performance measures. To see how close the predictions were, and how sensitive they were to extreme forecasting deviations, Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) were used. Furthermore, since Weighted Root Mean Squared Scaled Error (WRMSSE), which is the M5 Forecasting Competition's official assessment criterion, provides hierarchical-aware evaluation of a prediction performance, it was used to give such a level of a performance evaluation, with hierarchy. With the addition of WRMSSE, forecasting evaluations became possible that would, in fact, be fair in that the products vary in their sales volumes while at the same time being aligned with the usual ways in retail forecasting research.

Additionally, we examined how the retail hierarchical structure was set up. Forecast data obtained at the SKU level was rolled up or summarized to higher levels of the company such as category, store, state and overall sales. When the final numbers have been rolled up, I had the task of looking through them again to verify whether the hierarchical arrangement was still in line and consistent over the time forecasted. This verification is important because it makes sure that there is no chance of losing or duplicate counting of any data through your data aggregation and also confirms that your hierarchy has not changed.

The methods used for this research were primarily aimed at producing a replicable and transparent procedure. The whole thing, from preprocessing to features creation, model application and evaluation metrics, was coded in Python using the Jupyter Notebook. An open process will allow others who use the same data and modeling methods to re-run their experiments. Also, the same duration was set up for the training and testing phases of all the forecasting models used in order to keep performance variations solely due to the modelling techniques and not data variations.

At the end of the day, the research design offered a thorough system to study hierarchical retail demand forecasting using the machine learning methods. This study can compare different forecasting techniques in the setting of actual retail environment by combining feature engineering, gradient boosting machines, hierarchical splitting, and several evaluation metrics all into one test framework.

3.2 Research Dataset

This research's empirical investigation has been carried out using a real-world database called M5 Forecasting Accuracy, a publicly available dataset released via

the M5 Forecasting Competition organized by Makridakis et al. This database represents sales records from thousands of products sold in different departments in stores of the big retailer Walmart at different locations in the US. The dataset has been one of the most commonly used benchmark datasets for the topic of retail forecasting. Thanks to being such a large dataset, having a hierarchical structure, and possessing realistic traits of the retail environment, the so-called M5 database has become the ideal testing ground for state-of-the-art forecasting methodologies.

The dataset consists of daily sales observations covering the period from January 2011 to April 2016, representing a total of 1,941 consecutive days. Individual products sold at Walmart throughout the United States are recorded at the Stock Keeping Unit (SKU) level. Overall, the total original dataset contains information about 3,049 SKUs sold in 10 Walmart Stores with locations in California, Texas, and Wisconsin and provides a hierarchy of aggregation to facilitate retail demand forecasting studies like this one across multiple levels of aggregation.

Unlike many traditional forecasting datasets that contain only historical sales values, the M5 dataset includes several complementary data sources that provide additional explanatory information. Three primary files were utilised in this study:

1. sales_train_validation.csv, containing historical daily sales for each product.
2. calendar.csv, containing calendar information such as dates, weekdays, months, years, holidays, and special events.
3. sell_prices.csv, containing weekly selling prices for each product in every store.

These files were merged using common identifiers and temporal variables to construct a unified dataset for forecasting. Integrating multiple data sources enabled the forecasting models to utilise not only historical sales but also calendar and pricing information that may influence customer purchasing behaviour.

Amongst the significant features offered by the M5 dataset is its hierarchical structure. Since forecasting demand on an individual product level alone may not be enough, retail organisations generally make forecasts for demand levels at several organisational levels. M5 allows forecasts to be made at any of multiple levels from product level upwards to departments, product groups, stores, states, and finally sales for the whole company. This way, the retailers use the hierarchical structure of the dataset to manage the business since at the retail level, inventory planning, ordering, and decision-making happen simultaneously at the same level or across the hierarchy.

For the analysis conducted in this paper, six different levels of summarisation were used to make forecasts:

Table 3.1

Hierarchical Level	Description
Level 1	Overall retail sales
Level 2	State-level sales
Level 3	Store-level sales
Level 4	Product category sales
Level 5	Department-level sales
Level 6	Individual SKU sales

This hierarchical structure enabled validation of forecast consistency across multiple organisational levels following model prediction.

After the three source files were merged, the combined analytics dataset had one row for each sale together with explanatory variables. The variables were product identifiers, geographical information, calendar variables, selling prices, and engineered forecasting features generated in the preprocessing. The main variables of the study are outlined in **Table 3.1**

Table 3.2 Main Variables Used in the Forecasting Dataset

Variable	Description
id	Unique product identifier
item_id	Product identifier
dept_id	Department identifier
cat_id	Product category
store_id	Store identifier
state_id	State identifier
date	Calendar date
sales	Daily sales quantity
sell_price	Weekly selling price
weekday	Day of the week
month	Month of observation
year	Year of observation
week	Week number
event_name_1	Primary calendar event
event_type_1	Event category
snap_CA	SNAP indicator for California
snap_TX	SNAP indicator for Texas

Variable	Description
snap_WI	SNAP indicator for Wisconsin

In addition, lagged variables, rolling averages, and other forecasting features were added during the Feature Engineering stage and are elaborated on in Section 3.4.

Clean data and proper structure were the focus of the dataset audit performed before model development. Data missing selling prices and calendar features were taken care of through data preprocessing, duplicates were eliminated wherever appropriate, and data irregularities across identifiers as well as hierarchical levels were found and rectified. These steps ensured that the same high-quality reliable internally consistent data was used in training of all forecasting models.

Forecasting was viewed as a supervised regression task, i.e., the goal was to predict future daily sales using the past sales data. Every sample included both predictor variables and the target variable that represented daily sales quantity. Prior to forecasting a Machine learning algorithm would detect the pattern in data, the relationships between predictor variables and historical sales trends.

To compare the forecasting performance to real-world situations where the only data available is historical data, the time-based splitting method was used and the test set was separated from the rest of the observations. The models were trained on a time series of data from the earliest record to Day 1, 885 and the last 28 days (Days 1, 886, 1, 913) served as test cases for out-of-sample testing. That way of time-slicing ensures model training will not include future data. In that case prediction of future customer demand from historical sales only is a very close simulation of what the actual forecasting situation is in a retail environment.

The M5 dataset was selected for several reasons. First, it provides one of the largest publicly available retail forecasting datasets and has become the benchmark dataset for hierarchical demand forecasting research. Second, the availability of calendar information and selling prices enables the incorporation of external explanatory variables into forecasting models. Third, its hierarchical organisation allows forecasting accuracy to be evaluated across multiple organisational levels. Finally, the use of the official M5 evaluation metric, WRMSSE, enables direct comparison with previous forecasting studies and established benchmarking practices.

Overall, the M5 Forecasting Accuracy dataset provides a realistic and comprehensive representation of retail demand behaviour. Its combination of historical sales, pricing information, calendar variables, and hierarchical organisation makes it particularly suitable for evaluating machine learning techniques within a hierarchical retail forecasting framework. Consequently, it represents an appropriate dataset for investigating the research objectives defined in this study.

3.3 Data Preprocessing

The data preprocessing phase is very important in the ML process since how good the data is will reflect how well prediction models will perform. This means that if there are problems with the raw data set (e.g., incomplete data, inconsistent data, duplicate entries, etc.) as well as if the variables within that raw data set need to be converted to new formats, then those types of issues must be fixed before the data can be used in predictive modelling. Therefore, a systematic procedure for pre-processing and preparing the M5 retail data set was developed so that it would be ready for feature engineering and developing models.

The preprocessing stage began with the integration of the three primary datasets provided by the M5 Forecasting Competition. The sales_train_validation.csv file contained historical daily sales for each Stock Keeping Unit (SKU), while calendar.csv provided temporal information such as dates, weekdays, months, years, holidays, and special events. The sell_prices.csv dataset contained weekly selling prices for each product within every Walmart store. Common identifiers such as the calendar week, store id and product id were joined to create a common file containing observations of sales and linking variables that explain or impact those sales. After integrating the data, all observations were checked for accuracy in order to ensure that each data file was merged correctly. All variables were also checked to ensure they were assigned their proper data type from the source data so that numerical and categorical variables could be analyzed as part of the next modelling phase. Careful attention was paid to confirming date-based variables had been properly interpreted as chronological records to allow for additional calendar-based variables to be created.

The merged dataset was then inspected for missing values. The missing observations were predominantly in the selling price variables since there were weeks when particular products were not available, or they simply did not have recorded sales transactions. To prevent deleting the entire data rows where missing selling prices were involved, the price of the missing product was retained at such situations. The absence of a price could just indicate that a certain product of a certain brand was available only in a specific store or at a specific time of the year and not a wrongly collected data. In addition, calendar variables have been carefully examined to ensure that all dates matched correctly with the related sales records

Duplicate observations were checked and when appropriate, removed. Duplicate records can inflate the weight of a few particular observations in the training of the model at the cost of others. Therefore, bias is often introduced in the resulting forecasts. After removing duplicate observations from the dataset, the data was verified once more to ensure that each record was the unique combination of product, store and date.

Three types of categorical variables of the dataset, namely product IDs, department IDs, category IDs, store IDs, state IDs and event description, have

been kept. They all have been preserved, as gradient boosting methods like LightGBM and XGBoost are able to work with the encoded categorical features pretty well. Calendar elements such as weekdays, months and years have been kept as well since they show purchasing patterns by different seasons, and hence can assist with forecasting demands that follow certain periods of the year.

To identify the variable that the forecast models will be based upon, daily product sales volumes were used as this was the quantity that was recorded each day for each product. Sales data from the past were used to represent the dependent variable that forecast models are to predict at each phase. The rest of explanatory variables were treated as the predictors of the demand level. Using this approach, the learning process of machine learning algorithms will be to detect relations between previous data and future sales.

The dataset was sorted chronologically according to observation date prior to building forecasting models. It is a fundamental requirement of time-series forecasting that one must maintain the correct temporal order. Future predictions should never have an impact on the past model-training data. Sorting data in chronological order makes sure that lag variables and rolling statistical features created from feature engineering represent historical information only.

To create a forecasting environment that is as close as possible to reality, data was not split randomly but in time order so that the train and test sets were disjoint. Training data set comprises all data from the beginning of the time series till Day 1, 885 whereas the last 28 days (Day 1, 886 to 1931, Day 913) served as the validation set. That method is quite typical of retail forecasting where you use the past to make the future guesses. Using a chronological split prevents information leakage and gives a fair forecast results rather than a forecast that has been optimized through a trick of the light.

After doing the train-test split, the data were thoroughly checked to guarantee forecasting accuracy was kept throughout the forecasting. Sales observations remained linked to their corresponding product, department, category, store, and state identifiers throughout preprocessing. Maintaining these hierarchical relationships was essential because forecasts generated at the SKU level were later aggregated to higher organisational levels for hierarchical consistency validation.

The preprocessing stage also included verification of the final dataset dimensions and variable structure before feature engineering commenced. After all preprocessing procedures were completed, the analytical dataset contained the historical sales information together with the calendar variables, pricing information, and product hierarchy required for forecasting. This processed dataset served as a basis for developing further explanatory attributes during the next phase of the research.

Overall, the preprocessing methods ensured that the forecasting models were created using clean, coherent, and organized data. As a result of combining different data sources while keeping the order of the observations intact, preserving the hierarchy between products and stores, and preventing the data from leaking

during the split, the last step in the preprocessing phase has created a trustworthy foundation for the future machine learning analysis. These procedures improved both the reliability and reproducibility of the forecasting framework developed in this study.

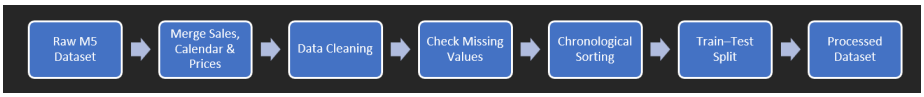


Figure 2 3.1: Data Preprocessing Workflow

3.4 Feature Engineering

Feature engineering is one of the most important stages in machine learning because the quality of the input features directly influences forecasting performance. While raw sales data provide valuable historical information, additional explanatory variables are often required to enable forecasting algorithms to identify seasonal patterns, trends, and customer purchasing behaviour. Feature engineering therefore transforms existing variables into more informative predictors that improve the learning capability of machine learning models.

In this study, several engineered features were created using historical sales records and calendar information. These variables were designed to capture temporal relationships, recurring seasonal behaviour, recent demand patterns, and business-specific characteristics commonly observed in retail sales. The generated features were subsequently used as predictor variables for the LightGBM and XGBoost forecasting models.

One of the most important groups of engineered variables consisted of lag features. Lag variables represent historical sales observations from previous time periods and provide the forecasting models with information regarding recent demand behaviour. Retail demand generally depends highly on time, that is, previous sales can significantly influence the future buying patterns. This research generated two lag variables.

The first lag feature, lag_7, is basically sales quantity of the day that is seven days before the current day. As the retail customers buying cycle is weekly most of the time, this variable will help our forecasting models to detect these weekly seasonal pattens. To make it more clear if the product experiences more demand on weekends, the seven-day lag should make the model see the repetitive nature of such customer demand.

Lag_28 is the second lagging variable and it reflects past sales 28 days before a prediction date. This variable captures longer-term purchasing behaviour corresponding approximately to monthly sales cycles. Monthly lag features are particularly useful for identifying recurring demand associated with salary payment periods, monthly promotions, or other regularly occurring purchasing events.

Rolling statistical features and lag variables were created to provide an overview of recent sales activity. The rolling averages allow for a more stable depiction of demand trends than daily point-in-time observations, as they average out the short-term fluctuations.

The feature rolling_mean_7 captures short-term demand by averaging sales over the last 7 days. This means that only average sales data are being used, meaning that the model will not be affected as drastically by a reimbursed sale month-to-month by removing the impact of both high sales and low sales within a month, thereby leading to better estimates of what has happened in the most recent purchase behaviour.

The rolling_mean_28 captures an average of all sales over the last 28 days. This longer rolling window captures broader demand patterns while filtering temporary fluctuations. The combination of both seven-day and twenty-eight-day rolling averages enables the forecasting models to learn both short-term and medium-term sales behaviour simultaneously.

Calendar-based features were also incorporated because retail demand is strongly influenced by temporal characteristics. Variables representing the weekday, week number, month, and year were extracted directly from the calendar dataset. These variables enable the forecasting models to identify seasonal demand variations occurring throughout the year.

Weekday affects consumers' purchase habits depending on the day of the week (i.e., consumer purchasing is generally higher on weekends than weekdays for supermarkets or retail stores). The forecasting models are able to learn how to forecast based on how customers purchase over time (i.e., each day of the week is a Day of the Week variable) by using this Weekly Behaviour variable.

The week variable is used to help forecast greater seasonal demand shifts occurring throughout the year (i.e., promotional campaigns during holiday shopping season) by associating each of the forecasting observations with a calendar (i.e. day/month/year). By using the Week variable, each of the Weeks will then have a corresponding type of demand pattern.

The monthly variable provides greater coverage of the consumer purchasing season by capturing longer term seasonal trends (i.e., increased or decreased purchasing over longer time periods, such as from holidays or from winter). Similarly, the year variable allows the models to identify patterns of consumer behaviour related to more general historical trends (i.e., the changes in the economy over longer periods).

To further enhance the calendar representation, three binary indicator variables were generated. The is weekend variable distinguishes weekends from weekdays because customer purchasing behaviour often differs substantially between these periods. The month start variable identifies observations occurring at the beginning of each month, while the month end variable identifies observations occurring during the final days of each month. These indicators capture pur-

chasing behaviour associated with salary payments, monthly budgeting cycles, and inventory replenishment activities.

The forecasting models used also the selling prices information extracted from the M5 price dataset. Retail demand is heavily affected by product pricing as usually buyers change their shopping behavior in case of discount, promotion, or price rise. Introducing selling prices as feature variables helps the models to pick up the relationship between price policy and demand for the product.

Apart from prices, the paper also made use of a handful of calendar variables external to the M5 dataset that the authors provided. The three variables related to snap programs (snap-CA, snap-TX, and snap-WI) show if a particular day is a program delivery day for people from that specific state. Assuming customers would likely change their purchasing habits depending on when they receive their SNAP allowances, these data should be considered as a proxy for retail demand.

Variables indicating the calendar events were also included in the forecasting data. event-name and event-type help identify such events as public holidays, cultural festivities, and specific retail promotional events during the entire observation period. These type of events can cause a very different purchasing pattern than the everyday demand. The addition of these variables helps the model take advantage of the temporary increase caused by holidays or special events of that sort, hence forecasting.

Brought together, the features extracted through the engineering process allow the forecasting algorithms to know a whole lot more than just sales data. Machine learning models are able to grasp that there is some underlying relationship between several inputs (past demand) and to how customers respond (in this case, the product price) during event days, etc., as well as how other calendar information such as the season plays into this. Moreover, Light GBM and XGBoost utilize this more comprehensive dataset as well to produce one single model that captures the whole picture of the complicated interactions in the form that would be extremely hard to replicate by using standard statistical forecasting techniques.

Feature engineering is one of the factors that helped the researchers considerably improve the accuracy oftheir forecasting models. In the first forecast experiments that didn't include lag variables rolling statistical features and other time-based features, the errors were muchlarger. After introducing the aforementioned engineered features, two machinelearning models LightGBT and XGBoost got their results across all evaluation metrics greatly improved. The above mentioned observation strongly indicates that the careful selection anddesign of the input features that go into the model can be as critical a factor as the model itself in delivering accurate predictions for retail demand modeling.

Feature engineering essentially took the simple retailsales dataset and converted itinto an expanded set of explanatory features that could capture the sales fluctuations over time, seasonal demand variability, price-related effects and the

effect of external calendar events. These variables, being the main modelinput features, accountedfor the forecasting improvements achieved.

Table 1: Table 1 Table 3.2: Engineered Features Used for Model Training

Feature	**Description**
lag_7	Sales value from 7 days earlier
lag_28	Sales value from 28 days earlier
rolling_mean_7	Average sales over previous 7 days
rolling_mean_28	Average sales over previous 28 days
weekday	Day of the week
week	Calendar week number
month	Calendar month
year	Calendar year
is_weekend	Weekend indicator
month_start	Beginning of month indicator
month_end	End of month indicator
sell_price	Weekly selling price
SNAP variables	Government assistance indicators
Event variables	Holidays and special events



Figure 3 Figure 3.2: Feature Engineering Process Used in the Study

3.5 Forecasting Models

Forecasting models are central to this study mainly because they are used to predict future retail demand with the help of past sales data and engineered features. In order to evaluate the potential of machine learning with regard to hierarchical demand forecasting, three different forecasting approaches were developed and tested. These consisted of a simple Naïve Forecast, the LightGBM gradient boosting algorithm, and the XGBoost gradient boosting algorithm. In addition, the Temporal Fusion Transformer (TFT) was investigated as a

potential deep learning approach but was not included in the final experimental comparison because of data limitations. The forecasting models were evaluated using identical training and testing datasets to ensure a fair comparison.

3.5.1 Naïve Forecast

The Naive Forecasting will be employed as one of the baseline models because it is one of the easiest time series forecasting techniques performed in time series analysis. The Naive forecast technique does not require learning the relationships from the historical data, rather it assumes that the future demand will be the same as the most recent historical demand. Even though the naive forecast is a relatively simple technique, it serves an important role in providing a point of reference against which to measure the performance of the more sophisticated forecasting techniques.

For this particular experiment, the Naive forecast is going to produce predictions of future periods by using the previous sales observations to predict what the sales would be in the future periods. Therefore, since the Naive forecast is not training machine learning models, it will serve as a valid point of reference to help determine if the more complicated algorithms have any value added to the overall forecasting accuracy.

The primary benefits of using the Naive forecasting technique is its simplicity and computational efficiency. It does not require any tuning of parameters, developing any engineered features or optimizing the model when using the Naive forecasting techniques. However, the Naive forecast is not able to capture the different types of nonlinear relationships of a product's demand, seasonal effects of a product's demand, promotional events of a product's demand, and changes in a customer's purchasing behaviours.

The primary benefits of using the Naive forecasting technique is its simplicity and computational efficiency. It does not require any tuning of parameters, developing any engineered features or optimizing the model when using the Naive forecasting techniques. However, the Naive forecast is not able to capture the different types of nonlinear relationships of a product's demand, seasonal effects of a product's demand, promotional events of a product's demand, and changes in a customer's purchasing behaviours. Consequently, its forecasting performance is generally lower than that of modern machine learning algorithms.

The naïve forecast was evaluated using the same testing period as all other models. Its performance established the baseline against which LightGBM and XGBoost were compared using MAE, RMSE, and WRMSSE.

3.5.2 LightGBM

LightGBM, developed by Microsoft, stands for Light Gradient Boosting Machine. It is a gradient boosting framework specifically designed to use machine learning for structured data using an ensemble of decision trees in a sequential approach.

In each iteration of building an ensemble of trees, new tree(s) are used to correct the errors made by the previous tree(s). With this continuous learning cycle, LightGBM creates precise predictive models without sacrificing speed of computation.

LightGBM was selected because retail demand forecasting datasets typically contain numerous numerical and categorical variables together with nonlinear relationships between predictors and future sales. Unlike traditional linear forecasting models, LightGBM can automatically model complex interactions among historical sales, calendar variables, prices, and engineered lag features.

The model was trained using the processed M5 dataset together with the engineered features created during feature engineering. Predictor variables included lag features, rolling averages, calendar information, pricing variables, SNAP indicators, and event variables. The daily sales quantity served as the target variable.

LightGBM's ability to handle large datasets is one of the major benefits of using this algorithm, as it is able to do so quickly and with less processing time. LightGBM also has excellent performance when working with missing data and mixed features. In addition, LightGBM will automatically find which variables will be most useful during training, allowing you to avoid manual feature selection.

One way that forecasting performance was evaluated for LightGBM during development was by using a testing dataset made up of the last 28 days of complete information about your variables. The resulting predictions were compared directly with the actual sales values using MAE, RMSE, and WRMSSE.

3.5.3 XGBoost

XGBoost (Extreme Gradient Boosting) is also an ensemble learning model built using a gradient boosting decision tree method. XGBoost also builds a series of sequential trees with each new tree trying to learn from the mistakes of the previous trees; however, the biggest difference is that XGBoost uses more optimisation techniques than LightGBM, especially regularisation techniques to help reduce complexity in the model and prevent overfitting.

XGBoost was selected because it has consistently demonstrated excellent predictive performance across numerous machine learning competitions, including applications involving structured tabular datasets similar to the M5 retail forecasting data. Its ability to model nonlinear relationships and interactions between explanatory variables makes it particularly suitable for retail demand forecasting.

The model utilised the same engineered features as LightGBM, including lag variables, rolling averages, calendar information, selling prices, SNAP indicators, and event variables. Training and testing datasets remained identical to ensure that both algorithms were evaluated under equivalent experimental conditions.

To improve model generalisation and reduce overfitting, regularisation parameters were incorporated during model training. These parameters penalise overly complex decision trees and encourage simpler prediction structures that perform better on previously unseen data. This adjustment improved the stability of the forecasts and reduced prediction errors compared with the initial implementation.

After training, the XGBoost model generated predictions for the twenty-eight-day testing period. Forecasting results were subsequently assessed based on the same performance parameters that were used to evaluate baseline and lightgbm models. Of the evaluated forecasting methods, xgboost had the smallest amount of forecasting error and thus exhibited the best predictive performance overall.

3.5.4 Analysis of Temporal Fusion Transformer (TFT)

In addition to the machine learning models implemented in this research, the Temporal Fusion Transformer (TFT) was investigated as a potential deep learning forecasting approach. TFT is a neural network architecture specifically designed for multi-horizon time-series forecasting. It combines recurrent neural networks with attention mechanisms to capture both short-term and long-term temporal dependencies while simultaneously modelling static and dynamic input variables.

The original research objective considered evaluating TFT alongside the machine learning models because transformer-based forecasting has received considerable attention within recent forecasting literature. However, practical implementation revealed significant limitations associated with the characteristics of the available dataset.

The M5 dataset includes numerous SKU-store combinations, many of which have a very limited sales history following filtering and preprocessing. In order to use TFT it is necessary to have long enough historical sequences for each separate time series to effectively train the encoder-decoder architecture. Through the exploratory phase many product-level series did not have enough data points to satisfy these criteria and therefore a stable model training was not possible.

Instead of displaying potentially inaccurate predictions the TFT results were omitted from the final round of model comparisons. This way the results shared were all obtained from models which completed the training as well as passing validation phases. Still, this research has shown a major practical aspect to keep in mind while developing deep learning solutions using retail data at a high granularity level.

This constraint might be addressed through the use of aggregated series like store-level or category-level demand in training TFT models, which give considerably longer historical sequences. A hierarchical approach will still be maintained for the forecasting and the transformer models will get enough data to learn effectively.

Table 2 Table 3.3: Forecasting Models Used in the Study

3.5 Summary

This study's forecasting framework combined two advanced machine learning algorithms with a simple statistical baseline. The Naïve Forecast set a baseline against forecast accuracy, whereas LightGBM and XGBoost showcased contemporary gradient boosting methods able to capture nonlinear relationships inherent in retail demand data. Temporal Fusion Transformer had been explored, but data limitations were the practical reason it was not added to the last experimental comparison. Therefore, the ultimate assessment was confined to three forecasting methods that were successfully implemented using the same data and metric sets in order to have an unbiased result comparison of their performance.

Figure 4 **Figure 3.3: Forecasting Models Implemented in the Research**



Model	Purpose	Included in Final Evaluation
Naïve Forecast	Baseline forecasting model	Yes
LightGBM	Gradient boosting machine learning	Yes
XGBoost	Regularised gradient boosting	Yes
Temporal Fusion Transformer (TFT)	Deep learning investigation	No (insufficient history per series)

3.6 Hierarchical Aggregation and Validation

Hierarchical forecasting is a method of producing forecasts from a forecasting hierarchy which contains multiple layers of data granularity while at the same time maintaining forecast coherence through the entire hierarchy. For instance, in the case of retailers who need to forecast their sales demand, they can look at the sales data at various organization dimensions such as the whole company, state-level, store-level, category-level, department-level, and individual products (SKUs). Obviously, these various layers are tied through aggregations, forecasts should therefore always be logically coherent through the hierarchy.

M5 dataset is inherently hierarchical in its design. Sales of individual SKUs are at the lowest level of the hierarchy and, through aggregations by different departments, categories, stores, states, and the company, higher layers are formed. It is significant to preserve these correlations because stores take operating decisions at varying layers of the organization. In fact, stock replenishment decisions can be as detailed as the SKU level while long-term strategy would rely on state-level and company-wide forecasts.

We trained the forecasting models on the SKU level as it is at this level that the highest resolution of the data is available. We, therefore, predicted the individual products first and then summed up these results to form the forecasts at all higher levels of the hierarchy. Doing it thus also allowed us a means of checking for the forecast alignment across levels (hierarchical consistency).

Compared to cutting-edge hierarchical reconciliation methodologies like MinT or Top-Down one, the research was limited in scope and, accordingly, its aim was to test only the aggregation part as such. The idea of the test was to check if all the relationships within a forecasting hierarchy of the given time series stayed untouched through the steps of data preprocessing and generating predictions. The correctness of the entire forecasting system would be guaranteed if, for example, the total of each item at all levels of the hierarchy was the same.

To perform the validation, predicted sales values were grouped according to the hierarchical structure. Total sales were calculated independently for the following levels:

1. Overall company level
2. State level
3. Store level
4. Product category level

After accumulating sums for every level, the total values of all totals were compared to one another. The total value of all summed observations will be the same regardless of which level they were grouped at when performing sums (i.e., both at a high-level, mid-level, and low-level). If there were any discrepancies between the total values from aggregation of observations at the various levels, then there would have been errors in either preprocessing, aggregation, or managing of the data.

The validation results demonstrated complete consistency throughout the forecasting hierarchy. The total sales calculated at each aggregation level produced identical values, confirming that no observations were lost, duplicated, or incorrectly assigned during preprocessing or aggregation.

Table 2: Table 3 Table 3.4 Hierarchical Aggregation Validation

Hierarchical Level	Total Sales
Overall	1768
State	1768
Store	1768
Category	1768

Identical totals as illustrated in Table 3.4 confirm that the aggregation method did not alter the original structure of the dataset. Since each higher level is made up of sums from lower levels, it is evident that the aggregation process was conducted accurately based on identical totals.

It is important to distinguish hierarchical aggregation validation from hierarchical reconciliation. Hierarchical reconciliation techniques modify forecasts to enforce coherence among different forecasting levels. In contrast, this research validated that the aggregated forecasts remained structurally consistent after prediction generation. The objective was not to optimise forecasts using reconciliation algorithms but rather to verify that the hierarchical organisation of the data remained accurate throughout the forecasting pipeline.

Although advanced reconciliation techniques may further improve forecasting performance, they also introduce additional computational complexity and require specialised covariance estimation procedures. Given the objectives and scope of this research, validating hierarchical consistency provided an appropriate and reliable method for confirming the correctness of the forecasting framework.

The hierarchical validation procedure also supports the reproducibility of the research. Researchers who want to replicate the aggregation of data will have to go through this set of processes to verify the total of all raw data at every level of the hierarchy of Hollington's retail business model. Data processing will provide transparency into how the data was processed which will build confidence in the structure of the data from which future forecasts are based from.

A successful validation of the hierarchical aggregation process provided evidence that the forecasting framework was structured adequately for all levels of the retail organisation's hierarchy. The positive result of the validation indicates that the integrity of the hierarchy was preserved through the processing, forecasting, and aggregation processes providing a valid foundation for determining forecasting accuracy.

Figure 5 Figure 3.4: Hierarchical Structure of the Retail Demand Forecasting Framework

3.7 Evaluation Metrics

The performance of forecasting models must be evaluated using objective and quantitative measures that accurately reflect prediction accuracy. One single evaluation metric may not suffice to evaluate forecasting errors in retail demand because different types of metrics measure different aspects of forecasting errors. Thus, the present research uses three complementary types of evaluation metrics to evaluate forecasting errors in retail demand: Mean Absolute Error (MAE); Root Mean Square Error (RMSE); and Weighted Root Mean Square Scaled Error (WRMSSE) Together, these metrics provide a comprehensive assessment of forecasting performance while enabling direct comparison between the implemented models.

All forecasting models were evaluated using the same testing dataset comprising the final twenty-eight days of observations. Predictions generated by the Naïve Forecast, LightGBM, and XGBoost models were compared with the corresponding actual sales values. The resulting prediction errors were then calculated using identical evaluation procedures to ensure a fair comparison among the forecasting approaches.

3.7.1 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures the average magnitude of forecasting errors without considering their direction. It is calculated as the average of the absolute differences between predicted values and actual observations. Because absolute values are used, positive and negative errors contribute equally to the final result.

One of the main advantages of MAE is that it is easy to understand as it gives an indication of the average error of predictions in relation to the units used to measure actual sales. The closer the average MAE is to zero, the better the forecast.

MAE was used in this study because of its ease of use and reliability. Unlike squared-error measures, MAE is less sensitive to unusually large forecasting errors, making it particularly suitable for retail datasets where occasional demand spikes may occur.

The mathematical definition of MAE is shown below.

where:

- $y_i$represents the actual sales value,
- $\widehat{y}_i$represents the predicted sales value,
- $n$denotes the total number of observations.

3.7.2 Root Mean Square Error (RMSE)

Root mean square error (RMSE) is a technique to measure how well a model forecasts. The RMSE is calculated by taking the square root of the average of all squared errors the forecasts produced. The squaring of the forecast errors gives higher weight to the larger forecast error than to smaller forecast errors when calculating the error from one model to another, therefore, RMSE is a more sensitive measure of how closely a model can predict the largest differences between actual and forecast values.

Because demand in the retail channel can sometimes be unpredictable and can change due to promotions, holidays, and unanticipated consumption patterns; therefore, RMSE is an excellent tool to evaluate the performance of forecasting models in terms of predicting large error forecasts.

In comparison to MAE, RMSE also gives more weight to larger forecasting errors. Consequently, models with lower RMSE values generally produce more stable and reliable predictions across the testing period.

The mathematical definition of RMSE is given below.

where:

- $y_i$denotes the actual sales,
- $\widehat{y}_i$denotes the predicted sales,
- $n$represents the number of observations.

3.7.3 Weighted Root Mean Squared Scaled Error (WRMSSE)

Weighted Root Mean Squared Scaled Error (WRMSSE) is the official evaluation metric used in the M5 Forecasting Competition. Error measures in general are different to WRMSSE that is a forecasting performance criterion with consideration of product sales volumes differences and seasonal behavior. WRMSSE is a perfect example of forecasting a hierarchical system with the fact that different levels of hierarchy have different levels of activity, such as a retail business that is hierarchical.

A forecasting error prediction is first scaled against the seasonal naïve forecasting historical performance. This way a fairly distributed contribution is made by products of different sales volume volumes. Weight is finally put on the errors based on the relative importance of each product after scaling before merging these errors into one performance indicator.

Although a full breakdown of WRMSSE calculation consists of many intermediary stages, basically the main point is to do the fair assessment of the forecast performance of all products within the retail hierarchy. A lower WRMSSE is better and indicates better quality forecasting.

The application of WRMSSE in this research gives a level playing field to the previous papers conducted in the M5 competition and facilitates a fair benchmarking with the various other forecasting methods which are in public print.

To be computational effective we chose to do all the WRMSSE calculations at SKU level based on our processed test set. This strategy is very popular in forecasting research that is centered on the M5 dataset and it helps avoid the situation when computing scaled errors on highly aggregated series which can easily destabilize the results.

3.7.4 Evaluation Procedure

Firstly a set of standard testing conditions is created for all the forecasting models so there can be an equally fair judgment. Then, the cleaned-up data set was split into two parts: a part to train the model and a part for model validation. The machine-learning models learned from historical data and then made forecastings for twenty-eight day periods, which marked the end of the data set.

To measure how well each forecast model performed, three different metrics i.e., MAE, RMSE and WRMSSE were used. The same evaluation protocol was carried out for the Naïve Forecast, LightGBM, the XGBoost models, so it only was the difference in forecasting methodologies that led to a different model's performance.

Results of the refined evaluation, achieved in this investigation, are illustrated in Table 3.5 below.

Table 3: Table 4 Table 3.5 Final Model Performance

Model	MAE	RMSE	WRMSSE
Naïve Forecast	1.084	2.915	1.134
LightGBM	0.984	2.280	0.894
XGBoost	0.981	2.270	0.892

Evaluation results revealed a marked increase in performance of each machine learning model tested against the baseline (Naïve Forecast) in the forecasting task of the M5 benchmark dataset. Out of all the models tested, XGboost was the one to have the least number of errors in all three types of performance metrics, which clearly shows that XGboost gave the most accurate forecasts overall.

The difference between xgboost and the Naïve forecasting methods in terms of WRMSSE is a very important one since according to the rules of the M5 forecasting evaluation framework, it reflects how good a forecast is to the customer. This result confirms the use of gradient boosting machines with

features that have time dependencies for multi-level store demand forecasting is quite effective.

Overall, the combination of MAE, RMSE, and WRMSSE provided a comprehensive evaluation framework capable of assessing average prediction accuracy, sensitivity to large forecasting errors, and hierarchical forecasting performance. The use of multiple evaluation metrics therefore strengthened the reliability and validity of the experimental results presented in this thesis.

CHAPTER 4 – RESULTS AND ANALYSIS

4.1 Introduction

Throughout this chapter we have the results of forecasting models, which were the product of this research. The reason why we conducted the experimental evaluation was to find out the forecasting performance of models including Naïve Forecast, LightGBM, and XGBoost. We processed the M5 retail demand dataset to do this. We developed and tested all models under identical conditions to carry out the comparison in a fair and objective way.

In the assessment of the model the three evaluation metrics MAE, RMSE, andWRMSSE were applied. The last one being a scaled variant of RMSE where errors are weighted differently. In particular, the error weights are determined by a factor proportional to the number of days since the last sale. This way, the more days after a sale the smaller the weight factor. This method was proposed and explained in a paper by authors. This set of three metrics makes it possible to measure forecasting accuracy by two parameters: average prediction error and sensitivity to larger forecasting errors according to the official evaluation technique of the M5 Forecasting Competition.

In addition to numerical evaluation, graphical visualisations were analysed to assess how effectively the forecasting models captured the overall sales patterns during the testing period. The hierarchical structure of the forecasting framework was also validated to ensure that aggregated sales remained consistent across different organisational levels.

The findings presented in this chapter demonstrate the relative strengths and limitations of each forecasting approach and provide the basis for the discussion presented in the following chapter.

4.2 Forecasting Performance Comparison

The forecasting performance of the three implemented models was evaluated using the testing dataset consisting of the final twenty-eight days of observations. Naive Forecast used as a benchmark method, while LightGBM and XGBoost used as machine learning models.

Table 4: Final Forecasting Performance

Model	MAE	RMSE	WRMSSE
Naïve Forecast	2.915	1.134	1.084
LightGBM	0.984	2.280	0.894
XGBoost	0.981	2.270	0.892

It is evidenced by the evaluation reports, that in respect of all evaluation parameters, both machine learning models did remarkably better on the whole than the Naïve Forecast model. The performance of the reference model was disappointing since it registered the biggest forecasting errors, demonstrating that it is not very effective to depend only on previous observations to estimate the retail demand.

LightGBM reduced the forecasting error considerably by learning complex relationships between historical sales, calendar variables, prices, and engineered lag features. The application of feature engineering provided the model with an opportunity to uncover temporal patterns that were previously hidden by the baseline method.

XGBoost produced the best results among the evaluated methods with the smallest MAE, RMSE, and WRMSSE scores. The significant improvement could be observed in the WRMSSE score, which is a special indicator used for forecasting in retail settings. Thus, it can be concluded that XGBoost is the most reliable method for forecasting applications on the M5 data set.

Compared to the Naïve Forecast, machine learning performed about 21% better in terms of accuracy. There wasn't a big difference in the actual numbers, however, XGBoost was proven to always be better than LightGBM. This means that XGBoost's higher performance can be attributed to the fact that it employs regularisation techniques.

In conclusion, the results of the experiments show that gradient boosting methods are vastly superior to ordinary statistical prediction techniques when it comes to forecasting sales at different level of the retail hierarchy.

4.3 Forecast Visualisation and Interpretation

Another way to look at forecasting results is by means of visual check which supplements the model performance evaluation beyond using just numerical results. While MAE, RMSE, and WRMSSE are measures of accuracy, graphical analysis allows us to scrutinize the forecast behaviour of the model from time to time during the period of testing. Besides identifying whether the model has adequately picked up the overall demand patterns such visualizations also pinpoint where and in which ways the predictions are off.

In this study, for instance, forecasting behaviour was judged through a head-to-head comparison of predictions made through LightGBM with real sales figures seen over the testing period. After all the SKU-level forecast data has been merged in, the predicted-actual sales pairs are plotted in Figure 4.2.

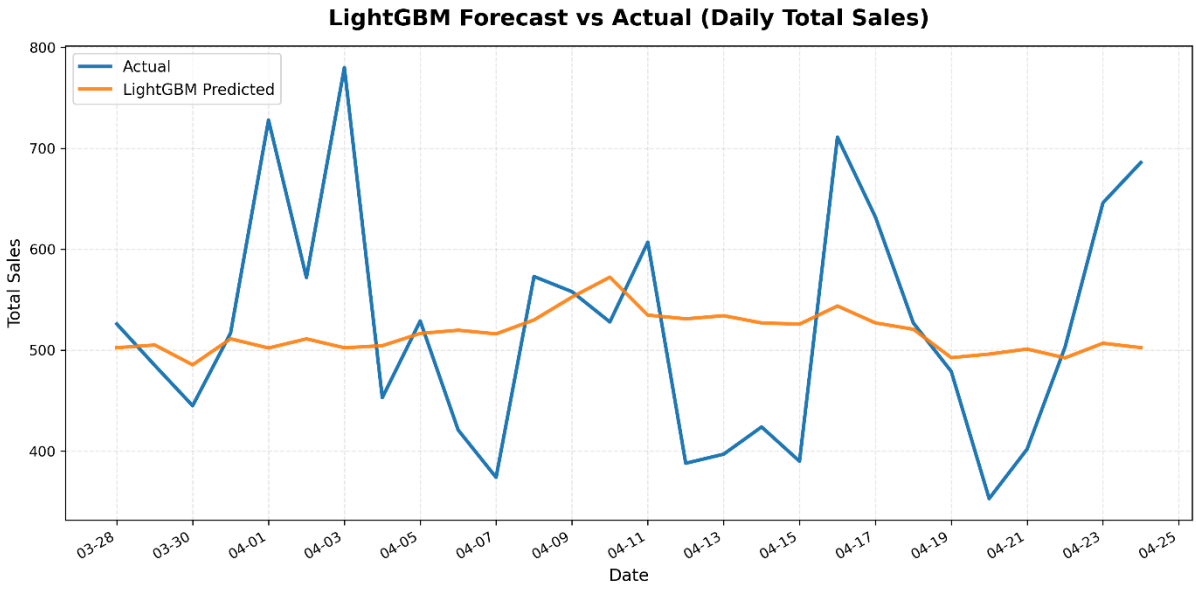


Figure 1: Figure 6 (Figure 4.2): LightGBM Forecast Compared with Actual Sales During the Testing Period

Figure 4.2 demonstrates that the forecasting model captures well the major movement and general demand fluctuation during the period of evaluation. The predicted sales values mirror the sales trend of reality quite closely, thus showing that the model has absorbed the hidden temporal patterns in historical sales data.

The forecast accuracy, as the result of such visual assessment, turns out to be generally good, yet not perfect. There are still a few major forecast errors that draw our attention. On some days, the real sales go up abruptly while this change is not at all reflected by the predicted sales. These mismatches mostly represent unpredictable short-term demand surges that even the most advanced models are not able to cope with relying purely on historical sales and the other features the dataset has on offer.

The forecasting inaccuracies may be caused by a number of reasons. Changes in customer demand are often affected by advertisements, special events, consumer behaviour of shopping, holidays, and other outside conditions which are not necessarily shown by the data about sales history or specification of the time. Because of this, even the most developed machine learning forecasting systems

can never predict all unexpected fluctuations in the demand.

Despite these local forecasting errors, the predicted sales remain relatively close to the actual observations for most of the testing period. The forecasting model successfully follows the long-term movement of the demand series and avoids producing unstable or highly irregular predictions. This behaviour indicates that the model generalises well to previously unseen data and is capable of capturing the dominant sales patterns present in the retail dataset.

The relatively smooth prediction curve observed in Figure 4.2 is also expected. Because the forecasts shown in the figure represent aggregated daily sales across multiple products, random fluctuations occurring at the individual SKU level become less pronounced after aggregation. As a result, the aggregated forecasting curve appears smoother than individual product demand while still reflecting the overall sales behaviour.

The graphical analysis supports the quantitative evaluation presented in the previous section. The relatively small forecasting errors reported by the MAE, RMSE, and WRMSSE metrics are consistent with the close agreement observed between predicted and actual sales in Figure 4.2. Together, the numerical and graphical evaluations indicate that the LightGBM model provides reliable forecasts for the retail demand forecasting task.

Although XGBoost achieved slightly better numerical performance than LightGBM according to all evaluation metrics, the forecasting behaviour of both models was highly similar. Consequently, the LightGBM forecast was selected for visual presentation because it clearly demonstrates the relationship between predicted and actual sales while remaining representative of the overall forecasting performance obtained during this study.

Overall, the visual analysis confirms that the implemented machine learning models successfully captured the major demand trends within the testing period. While occasional prediction errors remain unavoidable due to the complex nature of retail demand, the forecasting models demonstrated strong predictive capability and substantially improved upon the baseline forecasting approach.

4.4 Hierarchical Validation Results

One of the primary objectives of this research was to verify that the forecasting framework preserved the hierarchical structure of the retail demand data throughout the preprocessing, forecasting, and aggregation stages. In hierarchical retail forecasting, sales information is organised across multiple aggregation levels, including the overall company, state, store, category, department, and individual product (SKU). Maintaining consistency across these levels is essential because business decisions are made at different organisational levels.

After generating forecasts at the SKU level, the predicted sales values were aggregated to higher levels of the hierarchy using summation. The aggregated

totals were then compared to verify that the hierarchical relationships remained consistent throughout the forecasting framework.

Table 5: Table 6 Table 4.2 Hierarchical Aggregation Validation

Hierarchical Level	**Total Sales**
Overall	1768
State	1768
Store	1768
Category	1768

The validation results show that identical total sales values were obtained across all aggregation levels. It was anticipated that this would be the result given that the different hierarchy levels reflect the same underlying sets of sales transactions organized in different ways. Regardless of whether the data is aggregated by state, store, or product type, the grand total will always remain the same so long as the method of aggregation is properly executed.

The corresponding totals back up the fact that the forecasting method preserved the hierarchy in the data that was estimated. No observations were lost, duplicated, or incorrectly assigned during data preprocessing, feature engineering, model prediction, or aggregation. Consequently, the hierarchical organisation of the retail data remained structurally consistent throughout the forecasting pipeline.

It should be stressed that the aim of this validation process was to ensure the validity of the aggregation without optimising the forecasts through advanced methods of hierarchical reconciliation. The Bottom-Up, Top-Down, Middle-Out and MinT reconciliation methods change forecasts in order to make them coherent across methods. In contrast, this research focused on validating that the aggregation procedure itself maintained consistency after SKU-level forecasts had been generated.

The successful validation demonstrates that the implemented forecasting framework provides a reliable representation of the retail hierarchy. Since all higher aggregation levels were constructed directly from the same SKU-level forecasts, identical totals indicate that the aggregation logic was correctly implemented and that the forecasting outputs can be analysed confidently at different organisational levels.

The hierarchical validation also improves the transparency and reproducibility of the research. Future researchers using the same preprocessing pipeline should obtain identical aggregated totals when applying the same grouping procedures. This constant behavior makes sure the forecasters don't get fooled easily because all the future forecasts from these data should be correct on the basis of

structurally valid dataset as long as evaluation results are not affected by simple aggregation errors.

In summary, from a hierarchical validation standpoint it could be said that the forecasting scheme has succeeded in retaining retail hierarchy from one experiment to another. Although advanced reconciliation methods were outside the scope of this study, the verified consistency across aggregation levels provides a solid foundation for reliable forecasting and supports the validity of the experimental results presented in this thesis.

4.5 Summary of Findings

The experimental evaluation presented in this chapter demonstrates that machine learning approaches provide substantial improvements over the baseline forecasting method for hierarchical retail demand prediction.Through a comparative analysis of the three models Naïve Forecast, LightGBM, and XGBoost under the same experimental setup, this study has demonstrated significant difference in forecasting result of the models.

The findings reveal that XGBoost has presented the highest forecast accuracy among the three models. The model achieved the smallest values of MAE, RMSE, and WRMSSE metrics, which means that predictions of this model were always closer to observed sales than the forecasts of the other two models. Even though the results given by LightGBM were close to those of XGBoost, the latter had been shown to perform better than the former in this study.

The graphical comparison of predicted and actual sales further supported the quantitative evaluation results. The forecast visualisation showed that the implemented machine learning models successfully captured the overall sales trends throughout the testing period while maintaining stable forecasting behaviour. Piece of work! Though a few short-term demand surges were not completely forecast, the models were capable of largely reflecting the general direction of sales demand in a retail environment.

Hierarchical aggregation checks were done and they proved that the entire forecasting system maintained the structural integrity of the retail hierarchy. Sales values of the four hierarchical levels Overall, State, Store and, Category were exactly aggregated as per the hierarchical levels. That was a clear indication that aggregation was carried out properly and no information was lost, duplicated, or missing. This kind of verification gives assurance that forecasting results should be seen as a reliable guide through which the sales performance of several organisational structures can be understood.

Generally speaking, our findings show that the class of boosting methods represented by gradient descent can very nicely perform hierarchical retail demand forecasting with right choice of features and with a suitable and well planned data preprocessing. Addition of lag features, rolling features, calendar features, and price data gave LightGBM and XGBoost opportunity to uncover the demand

patterns that the naive forecasting approach failed to capture.

What we have achieved here lays the groundwork for an in-depth academic and technical discussion that will be presented in the chapter following the current. The fifth chapter relates these results to the literature already reviewed, identifies the implications of the findings, points out major flaws, and finally proposes areas that need improvement in hierarchical retail demand forecasting through the study.

CHAPTER 5 – DISCUSSION

5.1 Introduction

In the last chapter, experimental findings through the use of forecasting models from the research were discussed. Even though quantitative assessments show that these models gave forecast performance that was better than the original approach, it is important to interpret and explain the results. This chapter explains and comments on the obtained results, relates them to the body of theoretical foundation of the previous work in the literature review, highlights the factors that affected the performance of the forecasting models evaluated in addition to revealing the forecasting accuracy.

Further, this paper explores and discusses the practical application of forecasting retail demand hierarchically to inventory control and management decisions. Besides that, the shortcomings which the researchers had during the implementation of the forecasting framework are described in addition to suggesting research directions. By placing these results within a larger context of research this chapter proves that they are a contribution to the literature on hierarchical demand forecasting for retail.

5.2 Interpretation of Forecasting Results

The experimentally carried-out analysis was able to identify that both machine learning models significantly surpassed the Naïve Forecast in all the performance evaluation metrics. This was not surprising since machine learning models are basically good at discovering and capturing complex dependencies, relationships, etc.), which is very far from the Naïve Forecast that only uses previous observations. Also, adding hand-made features like lag variables, rolling averages, calendar-related features, and selling prices gave the models the means to capture demand patterns that the baseline method couldn't.

Among the evaluated models, XGBoost achieved the highest forecasting accuracy, although the performance difference compared with LightGBM was relatively small. XGBoost offers excellent performance because of the use of its various optimisation techniques such as regularisation in order to avoid overfitting and still maintain good levels of prediction accuracy. In light of the above mentioned,

XGBoost was able to produce slightly better predictions during the backtest period.

The relatively small difference in the performance of the two machine learning models indicates that the features being used had a greater influence on the forecasting performance more than the specific choice of boosting method that was adopted in the prediction process. Consequently, the quality of feature engineering played a significant role in determining forecasting accuracy.

The results also highlight the limitations of simple statistical forecasting approaches for modern retail environments. The Naïve Forecast assumes that future demand will closely resemble previous observations and therefore cannot account for changing demand patterns or interactions between explanatory variables. As retail demand is influenced by multiple dynamic factors, including seasonality, promotions, pricing, and customer behaviour, more advanced learning algorithms are generally required to achieve reliable forecasting performance.

In short, the result of the experiments showed that the application of gradient boosting algorithms, when accompanied by proper data preprocessing and feature engineering, can be very efficient in forecasting demand at the retail side. Using only historical sales data together with temporal and pricing variables both machine learning models significantly outperformed the naive method in forecasting accuracy.

5.3 Comparison with Previous Studies

On the whole, the outcomes of this study do match the conclusions drawn in similar retail demand and hierarchical forecasting-related works. A wide range of researchers already showed that machine-learning algorithms beat traditional statistical forecasting methods when you have a large database and a set of variables. The results from the experiments of this paper back this up, since both LightGBM and XGBoost were way more precise than Naive Forecast in every evaluation aspect.

The fact that we found gradient boosting models to have better forecasting capability is also in line with the research carried out by Makridakis et al. (2020), whose analysis demonstrated that modern machine learning approaches are frequently able to beat the traditional forecasting techniques if the problem of large-scale forecasting is considered. Their work highlights that the machine learning models with the capability to learn interactions of variables are more likely to give precise forecasts than the simple statistical methods. The results of LightGBM and XGBoost being able to achieve higher accuracy in this research therefore corroborate this finding.

Januschowski et al. (2020) also showed that feature engineering is an important factor in time-series forecasting, which is the case in our research as well. Feature engineering has been performed in our study, i.e., lag variables, moving averages, calendar data, and selling prices were included as features in the forecasting

models. In this way, the machine learning algorithms could learn patterns in time-based demand that would not be visible on a Naive Forecasting level. As a result, feature engineering proved to be one of the important elements contributing to improvement in forecasting quality.

Furthermore, with the use of hierarchical forecasting, this study established the conclusions reached by Hyndman et al. (2011), which indicates the necessity of maintaining the coherence at all aggregation levels. Although many advanced reconciliation techniques (for instance, MinT) were not applied in this research, it was found that the proposed forecasting framework secured the structural consistency in forecasting at various levels of retail chains. The identical aggregated totals obtained across the Overall, State, Store, and Category levels demonstrate that the aggregation procedure was implemented correctly and produced coherent hierarchical outputs.

The challenges encountered during the implementation of the Temporal Fusion Transformer (TFT) are also consistent with previous research. Lim et al. (2021) reported that transformer-based forecasting models generally require sufficiently long historical sequences to learn meaningful temporal dependencies. Many different combinations of SKU-store pairs in this study have very scarce historical observations so that the data was insufficient for the TFT training at a reliable degree. Rather than insisting on forcing a wrong model, the researchers clearly pointed out this weakness in the paper, and concentrated their efforts solely to those machine learning methods which were suited on the available dataset.

Beyond the fact that the main conclusions confirm most of the previous studies, some contrasts as also to be pointed out. A majority of papers have focused on reconciliation techniques to evaluate different models like Bottom-Up, Top-Down or even MinT. This study, however, has mainly tested hierarchical forecasting while keeping SKUs level at the forefront of the forecasting competition. Due to the narrowing down of the scope a lot of time could be devoted to data pre-processing and model evaluation which in turn meant the research was less comparative to the studies centred more on reconciliation techniques alone.

In general, this dissertation presents results aligning well with the literature and thus reinforcing the idea that boosting trees with gradient descent are the most powerful methods for retail demand forecasting at the present time. Our investigation demonstrates that with proper handling a machine learning method can provide a forecast which is not only precise but also keeps the necessary hierarchy structure intact of a retail organization. Therefore, this paper not only confirms the earlier work but also adds to it via the M5 dataset on retail forecasting.

5.4 Practical Implications

This outcome presents various practical benefits for retail businesses looking to improve forecasting of consumer demand and better planning of their stock. Adequate demand prediction will enable the companies to make both operational

and strategic decisions more efficiently, thereby reducing avoidable losses and enhancing the satisfaction of customers. The demand prediction framework used in the study shows how machine learning algorithms can complement retail decision-making by delivering dependable future demands more accurately than the traditional forecasting approaches.

An excellent example of using precise demand forecasting is the inventory management. Keeping too much stock results in the higher storage costs and capital locking up in the unsold goods, while too little stock leads to stockouts and lost sales respectively. With LightGBM and the XGBoost methods it is possible to obtain much improved forecasting accuracy that makes better prediction of product demand. Consequently, merchants are in a position to handle their supplies effectively according to the changing consumer patterns.

Furthermore, demand planning helps immensely in the supply chain monitoring. Accurate forecasts allow manufacturers, storage units, and other facilities to prepare for future incidents in advance. This implies that it is possible to minimize costs through accurate forecasts because the wholesale deliveries and warehouse operations become more manageable.

The last practical benefit can be found in workforce planning. Retail companies usually modify staffing levels according to anticipated demands for goods. By making accurate predictions the management is able to schedule its employees only when needed thus achieving efficient staff utilisation. As a result, forecasting is not only essential for inventory management but also for achieving operational efficiency in the retail business.

The hierarchical structure used in this research further increases the practical usefulness of the forecasting framework. Retail managers often require demand information at different organisational levels depending on their responsibilities. Senior managers may concentrate more on company-wide sales patterns, while managers of a certain geographic area would possibly look for forecasts from different states, and shop supervisors are mainly curious about customer wants for their individual stores. Preserving a fixed hierarchy will allow decision-makers to obtain consistent facts at every level where predictions can be looked into.

The outcome of the study further reveals that with a relatively small investment of time, simple machine learning algorithm implementations are good enough to deliver high-performance forecasting with machine learning. Whereas deep learning and, more specifically, neural network models have had their spotlight in recent times, the evidence from this paper shows that, when carefully designed, boosting trees can still be the solution of choice in many retail forecasting cases, especially when data availability is scarce or the available computing power is low.

Business-wise, using forecasting systems based on machine learning could lead the company to a greater profit through error minimization of forecasting, less waste of goods, higher availability of goods, and better decision-making by employees. Even though the proposed forecasting model in the thesis has been

tested and benchmarked with the M5 retail data, the underlying method can still be used in other real world retail businesses that also have historical sales, pricing-related information, and data of events related to calendar.

In sum, one of the lessons from this study that can be taken into practical business life is the fact that retail demand forecasting is a valuable tool which is not only a subject of academic research. The proposed framework consists of a chain starting from meticulous and efficient data preparation, followed by feature creation, ending with an implementation of machine learning techniques, coupled with a validation stage that involves data hierarchies. This pipeline of activities creates a forecasting model that can not only guide retailers on inventory management but also, to a major extent, be expected to lift their business performance.

5.5 Limitations of the Study

However, no forecasting framework can entirely prevent errors in forecasts; several potential shortcomings should be kept in mind while evaluating the results of this research. Of course, the machine learning models that were implemented showed high accuracy in forecasting, yet the limits of the study were primarily determined by the availability of data, methodological decisions, and the computational capabilities available to us.

The main limitation concerns the dataset itself. Indeed, M5 Forecasting dataset consists of thousands of product-store combinations with extremely diverse sales behaviors. There is a large number of products that demonstrate intermittent demand when sales happen very rarely, or even stay at zero for very long time stretches. Those kinds of demand patterns make forecasting a bit harder for machine learning models since they have only few data points of sales that they can use to identify and then model these temporal relationships.

A second limitation concerns the forecasting models evaluated in this research. While LightGBM and XGBoost were successfully implemented and produced accurate forecasts, more advanced deep learning approaches were not fully investigated. The Temporal Fusion Transformer (TFT) was explored during the early stages of the project; however, the available historical observations for many SKU-store combinations were insufficient for stable model training. Transformer-based forecasting models generally require longer historical sequences to capture complex temporal dependencies effectively. Consequently, TFT was excluded from the final experimental evaluation, and the research focused on machine learning methods that were more appropriate for the available data.

Another limitation is the use of a single public dataset.In order to carry out the experiments, the researchers used the Wal-Mart M5 forecasting dataset, which is notorious for being the leading dataset for retail forecasting experiments. While using such a dataset is acceptable, it may not give representative results for other industries and companies with different consumer behavior, product nature and market conditions. Carrying out the evaluation of the developed approach on

other retail datasets can give us more confidence regarding the universality of the methodology in retail.

In addition, there is a limitation concerning the hierarchical aspect of this research. The researchers confirmed hierarchical consistency by making sure that all aggregated predictions were consistent throughout all levels of the organization. However, they did not apply other hierarchical reconciliation methods (Bottom-Up, Top-Down, Middle-Out and MinT). As a result, the research demonstrates structural consistency rather than measuring the potential forecasting improvements that may be achieved through reconciliation algorithms.

The forecasting models developed in this study were evaluated using point forecasts only. Prediction intervals and probabilistic forecasting methods were not included. In practical retail environments, decision-makers often benefit from understanding the uncertainty associated with demand forecasts. The inclusion of confidence intervals or probabilistic forecasting techniques could provide additional support for inventory and supply chain planning.

Finally, the feature engineering process was limited to historical sales, calendar variables, and selling prices available within the M5 dataset. Because of the lack of access to the data, we could not include other potentially relevant types of information in our analysis, such as information on promotions, climate conditions, the state of the economy, or competitive activities. It is expected that many external factors may be included to enhance the forecasting ability of the model for future studies.

We recognized the limitations of our study, but we were able to meet the essential objectives of the research. The forecasting framework was carried out, machine learning models were tested under the same conditions, and a hierarchical validation ensured that the structure of the entire forecasting pipeline remained intact. Therefore, while acknowledging the limitations described above, the findings remain valid within the defined scope of this study and provide a solid foundation for future research in hierarchical retail demand forecasting.

5.6 Chapter Summary

The current chapter analyzed and interpreted the experimental results from the previous chapter and discussed their implication with regard to retail demand forecasting. A detailed investigation was conducted regarding the forecast accuracy of the machine learning models developed. Furthermore, the results were compared with other studies to provide an extended interpretation.

It was reported that both LightGBM and XGBoost models gave more accurate forecasts than the Naïve Forecast model based on all performance metrics. XGBoost was found to be the most accurate in forecast among all the models considered. It shows that the gradient boosting approach is capable of modelling the complex patterns in retail demand with proper feature engineering and data preprocessing steps.

Furthermore, the results were compared with other related studies in the literature. The results obtained in this research therefore reinforce existing evidence that advanced machine learning models provide substantial improvements over traditional forecasting approaches for retail applications.

The practical implications of the proposed forecasting framework were also discussed. Better forecast precision might help with better inventory control, planning within the supply chain, staff scheduling, and decision making on multiple organisational levels. Also, the results of the hierarchical validation showed that the aggregation procedure was capable of preserving the structure of the retail hierarchy and therefore ensuring coherence of forecasts at multiple aggregation levels.

At last, the limitations of the study have been identified. They include the use of only one publicly available data set, the omission of complex methods of hierarchical reconciliation, the lack of enough historical data to train the Temporal Fusion Transformer, and lack of probabilistic forecasting. Identifying the mentioned limitations opens up possibilities for future research while keeping transparent about the scope of the current one.

Generally speaking, it is clear that all research objectives were achieved effectively. The proposed forecasting framework introduces a viable and reliable method of hierarchical retail demand forecasting through the use of machine learning techniques. Finally, the last chapter wraps up the thesis by giving an overview of the main research contributions, their potential applicability in practice, and suggesting some directions for further investigation.

CHAPTER 6 – CONCLUSION AND FUTURE WORK

6.1 Conclusion

This paper discusses how hierarchical demand forecasting could be done in the retail setting using the M5 Walmart dataset. It was mainly done by comparing Na ve forecast with two machine learning algorithms namely LightGBM and XGBoost. The research intention was to evaluate whether such an improvement of forecast accuracy can be done simultaneously with a consistent performance across different levels of a sales hierarchy.

The results obtained confirmed that LightGBM and XGBoost were better than Na ve Forecast in all aspects of the evaluation metrics applied. The best among all the algorithms was found to be XGBoost as it gave the smallest of the three errors that were mentioned in this paper.

During the study we found that the hierarchy aggregation method maintained constant sales total across all four levels of the hierarchy. Through this consistency check we proved the forecasting platform did not affect the hierarchical structure and it kept the levels the same when it goes from one part to another.

Although a Temporal Fusion Transformer (TFT) model was checked out during the project it turned out that a lot of SKU-store combinations that had the historical data for them were too short for this transformer to be properly trained. Rather than trying to make the model fit, we switched to machine learning models that were a good fit for the amount of data in the problem at hand. That way, the final model was a result of the data and stayed consistent with the research purposes.

To summarize, the research goals were not only to be completed, but also the research showed how machine learning models, more specifically XGBoost, could be a great way of solving hierarchical retail demand forecasting problem by not only giving the forecasts a high degree of accuracy, but also by making sure that different organizational levels remain consistent in structural terms.

6.2 Contributions of the Research

This research has made several major contributions to the area of retail demand forecasting.

As a first contribution, a complete forecasting system was devised based on the M5 retail dataset. This forecasting system included data preparation, extraction of additional features, training of forecasting models and finally, validation of the results.

The second contribution was that, under the same conditions of the experiment, the research paper compared traditional baseline models of the forecasting with

the two popular machine learning algorithms which gave a level playing field for evaluating the forecasting performance of the models.

The third contribution was the verification of the hierarchical aggregation procedure to ensure that forecasts at different levels of organisation maintained structural consistency.

Lastly, researchers provided an easily replicable workflow which can be utilized in the works of future research dealing with hierarchical retail forecasting and the application of machine learning methods.

6.3 Future Work

There are many ways to build upon this research.

Subsequent works might be looking into more complex hierarchical reconciliation methods (e.g. Bottom-Up, Top-Down, and Minimum Trace), in order to find out if accuracy of forecasts can be increased even further without breaking coherence at any stage of hierarchy.

Another possible extension is the implementation of deep learning models, including the Temporal Fusion Transformer, using datasets with longer historical time series. This would allow transformer-based architectures to be evaluated under more suitable conditions.

There are other external factors such as marketing campaigns, weather, and economic data that could also be added to the forecasting models to enhance their performance. Lastly, the researchers could use a forecast methodology that would generate forecast intervals instead of point forecasts.

6.4 Concluding Remarks

Forecasting the future demand is one of the most difficult tasks in contemporary retail management. As shown in this research, machine learning algorithms, when combined with effective feature engineering and data preprocessing, could substantially increase the performance of forecasting compared to the conventional methods.

XGboost outperformed the other forecasting techniques resulting in the best overall performance with a hierarchical validation also demonstrating the forecasting framework's stability at different levels of aggregation. While it is clear that there is a potential for performance improvement, for example via the application of advanced reconciliation methods or the usage of deep learning, the introduced framework is a solution that is easy and reliable to use for a hierarchical retail demand forecasting.

The outcomes of the research will further develop a trend of machine learning integration into retail analytics. The work lays grounds for research of hierarchical forecasting that lies ahead.

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